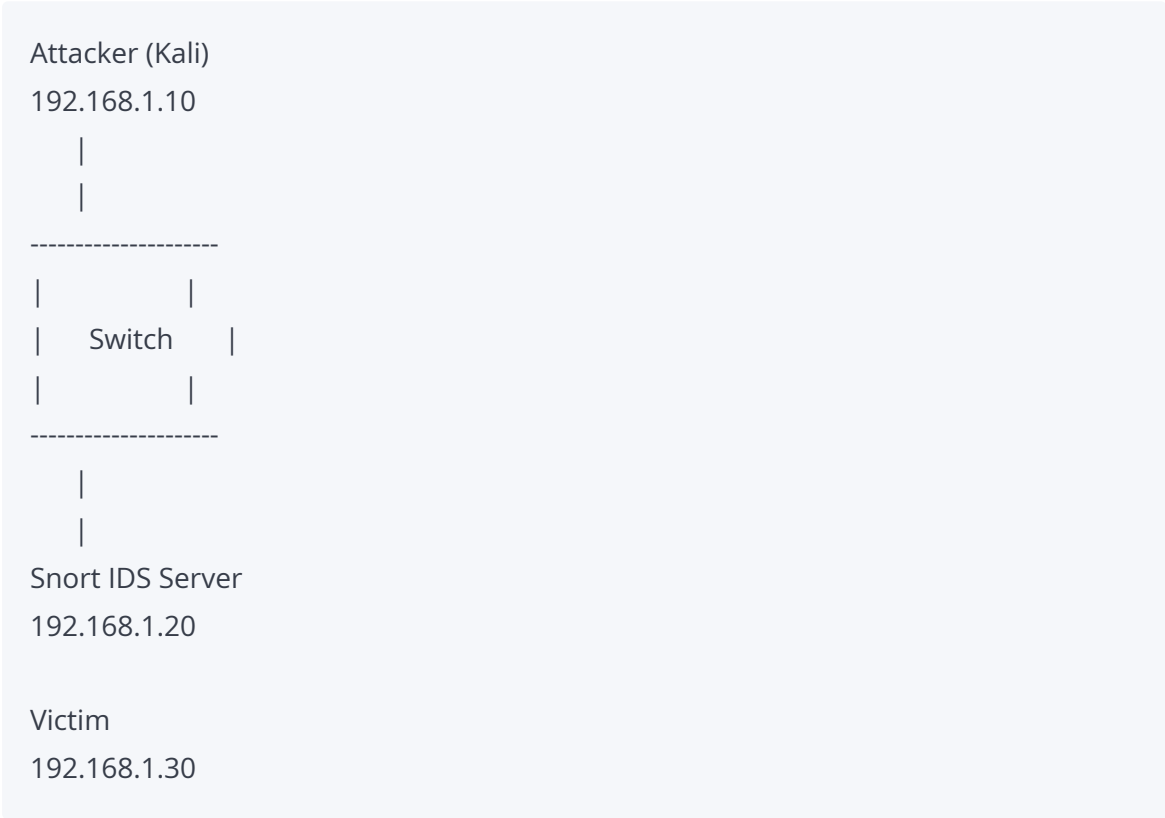


Snort IDS Full Lab

Lab Objective

Install and configure **Snort** as an IDS, create custom rules, generate attacks, and observe alerts.

Lab Topology



Step 1: Install Ubuntu Server

Use Ubuntu 22.04 or 24.04.

Check IP:

```
ip a
```

Example:

```
eth0: 192.168.1.20
```

Step 2: Update System

```
sudo apt update  
sudo apt upgrade -y
```

Step 3: Install Snort

```
sudo apt install snort -y
```

During installation:

```
HOME_NET:  
192.168.1.0/24
```

Enter your network range.

Step 4: Verify Installation

```
snort -V
```

Expected:

```
Version 2.9.x
```

Step 5: Find Network Interface

```
ip a
```

Example:

```
eth0
```

Remember the interface name.

Step 6: Test Snort Packet Capture

Run:

```
sudo snort -v -i eth0
```

Generate traffic from another machine:

```
ping 192.168.1.20
```

Output:

```
ICMP Echo Request  
ICMP Echo Reply
```

Press:

```
Ctrl + C
```

Step 7: Test Sniffer Mode

```
sudo snort -vd -i eth0
```

Shows:

```
IP Header  
TCP Header  
ICMP Header
```

Step 8: Test Full Packet Inspection

```
sudo snort -dev -i eth0
```

You will see:

```
Packet Payload  
Headers  
Hex Dump
```

Step 9: Configure Local Rules

Open:

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert icmp any any -> any any (msg:"PING DETECTED"; sid:1000001; rev:1;)
```

Save.

Step 10: Verify Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

Expected:

```
Snort successfully validated the configuration
```

Step 11: Run Snort IDS

```
sudo snort -A console \  
-c /etc/snort/snort.conf \  
-i eth0
```

Leave this terminal running.

Step 12: Generate Ping Attack

From Kali:

```
ping 192.168.1.20
```

Alert:

```
[**]  
PING DETECTED  
[**]
```

Step 13: Detect Nmap Scan

Edit:

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert tcp any any -> any any \  
(flags:S; msg:"PORT SCAN DETECTED"; sid:1000002; rev:1;)
```

Restart Snort.

Run from Kali:

```
nmap -sS 192.168.1.20
```

Alert:

```
PORT SCAN DETECTED
```

Step 14: Detect FTP Traffic

Rule:

```
alert tcp any any -> any 21 \  
(msg:"FTP CONNECTION"; sid:1000003; rev:1;)
```

Restart Snort.

From Kali:

```
ftp 192.168.1.20
```

Alert generated.

Step 15: Detect Telnet

Rule:

```
alert tcp any any -> any 23 \  
(msg:"TELNET DETECTED"; sid:1000004; rev:1;)
```

Step 16: Detect SSH

Rule:

```
alert tcp any any -> any 22 \  
(msg:"SSH LOGIN ATTEMPT"; sid:1000005; rev:1;)
```

Test:

```
ssh user@192.168.1.20
```

Step 17: Detect HTTP Access

Rule:

```
alert tcp any any -> any 80 \  
(msg:"HTTP ACCESS"; sid:1000006; rev:1;)
```

Test:

```
curl http://192.168.1.30
```

Step 18: Detect SQL Injection

Rule:

```
alert tcp any any -> any 80 \  
(msg:"SQL INJECTION";  
content:"UNION SELECT";  
sid:1000007; rev:1;)
```

Test:

```
curl "http://192.168.1.30/test.php?id=1 UNION SELECT 1,2"
```

Alert generated.

Step 19: Detect Password Attack

From Kali:

```
hydra -l root -P passwords.txt ssh://192.168.1.30
```

Snort detects repeated connections.

Step 20: Log Alerts

Create log directory:

```
sudo mkdir -p /var/log/snort
```

Run:

```
sudo snort \  
-c /etc/snort/snort.conf \  
-l /var/log/snort \  
-i eth0
```

View logs:


```
ls /var/log/snort
```

Step 21: View Alerts

```
sudo cat /var/log/snort/alert
```

Example:

```
PING DETECTED
PORT SCAN DETECTED
SSH LOGIN ATTEMPT
```

Important Snort Modes

Mode	Command
Sniffer	<code>snort -v</code>
Packet Dump	<code>snort -vd</code>
Full Packet Inspection	<code>snort -dev</code>
IDS Mode	<code>snort -A console -c snort.conf</code>

What is Snort?

Open-source Network IDS/IPS used to detect malicious traffic.

What are Snort Components?

- Packet Decoder
- Preprocessors
- Detection Engine
- Logging & Alerting System

- Output Module

What is SID?

Signature ID used to uniquely identify a Snort rule.

Example:

```
sid:1000001
```

Difference between IDS and IPS?

IDS = Detect + Alert

IPS = Detect + Alert + Block

What is HOME_NET?

Your internal network range.

Example:

```
HOME_NET = 192.168.1.0/24
```

This complete lab covers installation, rule creation, packet monitoring, ping detection, port-scan detection, SSH/FTP/HTTP monitoring, SQL-injection detection, and alert logging—enough for most CDAC, university, and interview practicals.